10

Bivariate Measures of Association

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Key Concepts

1. Scatterplot
2. Strength and direction
3. Covariance
4. Correlation coefficient
5. Fisher's *Z* transformation
6. Linearity assumption, causation, and restriction of range issues

We have considered various inferential tests in the last four chapters, specifically those that deal with tests of means, proportions, and variances. In this chapter we examine measures of association as well as inferences involving measures of association. Methods for directly determining the relationship among two variables are known as **bivariate analysis**, rather than **univariate analysis**, which is only concerned with a single variable. The indices used to directly describe the relationship among two variables are known as *correlation coefficients* (in the old days known as co-relation) or as *measures of association*.

These measures of association allow us to determine how two variables are related to one another and can be useful in two applications: (a) as a descriptive statistic by itself and (b) as an inferential test. First, a researcher may want to compute a correlation coefficient for its own sake, simply to tell the researcher precisely how two variables are related or associated. For example, we may want to determine whether there is a relationship between the GRE-Quantitative Reasoning (GRE-Q) subtest and performance on a statistics exam. Do students who score relatively high on the GRE-Q perform higher on a statistics exam than do students who score relatively low on the GRE-Q? In other words, as scores increase on the GRE-Q, do they also correspondingly increase their performance on a statistics exam?

Second, we may want to use an inferential test to assess whether (a) a correlation is significantly different from zero or (b) two correlations are significantly different from one another. For example, is the correlation between GRE-Q and statistics exam performance significantly different from zero? As a second example, is the correlation between GRE-Q and statistics exam performance the same for younger students as it is for older students?

The following topics are covered in this chapter: scatterplot; covariance; Pearson product-moment correlation coefficient; inferences about the Pearson product-moment correlation coefficient; some issues regarding correlations; other measures of association; SPSS and **R**; and power. We utilize some of the basic concepts previously covered in Chapters 6 through 9. New concepts to be discussed include the following: scatterplot, strength and direction, covariance, correlation coefficient, Fisher's Z transformation, linearity assumption, causation, and restriction of range issues. Our objectives are that by the end of this chapter, you will be able to (a) understand the concepts underlying the correlation coefficient and correlation inferential tests, (b) select the appropriate type of correlation, and (c) determine and interpret the appropriate correlation and inferential test.

10.1 What Bivariate Measures of Association Are and How They Work

Challie Lenge, along with her accomplished cohort of graduate research assistants working in the statistics lab, continues to assist with various research projects. We now find her embarking on exciting challenge with a community partner.

The faculty advisor for the stats lab received a telephone call from Dr. Amberly, the director of marketing for the local animal shelter. Based on a recent survey of donors to the shelter, it appears that the donors who contribute the largest donations also have children and pets. In an effort to attract more donors to the animal shelter, Dr. Amberly is targeting select groups—one of which she believes may be families that have children at home and who also have pets. Dr. Amberly believes if there is a relationship between these variables, she can more easily reach the intended audience with her

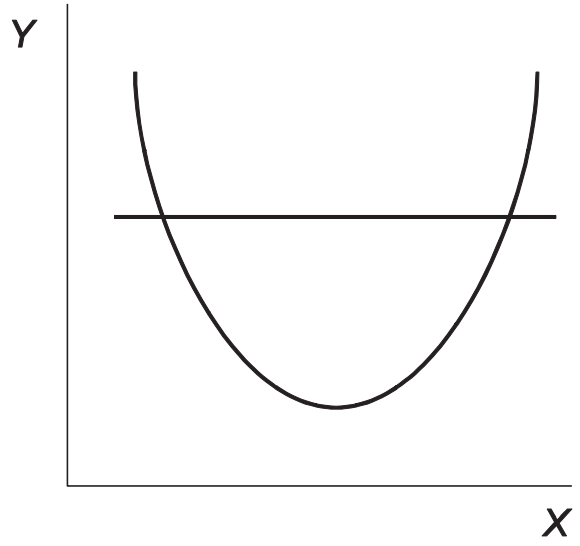


FIGURE 10.4
Nonlinear relationship.

10.2 Computing Bivariate Measures of Association Using SPSS

Next let us see what SPSS has to offer in terms of measures of association using the children–pets example dataset. SPSS has two tools for obtaining measures of association, dependent on the measurement scale of your variables: the Bivariate Correlation program (for computing the Pearson, Spearman's rho, and Kendall's tau) and the Crosstabs program (for computing the Pearson, Spearman's rho, Kendall's tau, phi, Cramer's phi, and several other types of measures of association).

10.2.1. Bivariate Correlations

Step 1. To locate the Bivariate Correlations program, we go to "Analyze" in the top pull-down menu, then select "Correlate," and then "Bivariate." Following the screenshot of Step 1 in Figure 10.5 produces the "Bivariate" dialog box.

Step 2. Next, from the main "Bivariate Correlations" dialog box, click the variables to correlate (i.e., "Number of children" and "Number of pets") and move them into the "Variables" box by clicking the arrow button. In the bottom half of this dialog box options are available for selecting the type of correlation (*this is where it's important that you understand the measurement scales of your variables so that you are computing the correct correlation coefficient given the scale of measurement of your variables*), one- or two-tailed test (i.e., directional or nondirectional test), and whether to flag statistically significant correlations. For illustrative purposes, we will place a checkmark to generate the "Pearson," "Kendall's tau-b," and "Spearman's rho" correlation coefficients. We will also select the radio button for a "Two-tailed" test of significance and at the very bottom check "Flag significant correlations" (which simply means an asterisk will be placed next to significant correlations in the output). See the screenshot of Step 2 in Figure 10.6.

10.3 Computing Bivariate Measures of Association Using R

Next we consider **R** for a bivariate measures of association model. Note that the scripts are provided within the blocks with additional annotation to assist in understanding how the command works. Should you want to write reminder notes and annotation to yourself as you write the commands in **R** (and we highly encourage doing so), remember that any text that follows a hashtag (i.e., #) is annotation only and not part of the **R** script. Thus, you can write annotations directly into **R** with hashtags. We encourage this practice so that when you call up the commands in the future, you'll understand what the various lines of code are doing. You may think you'll remember what you did. However, trust us. There is a good chance that you won't. Thus, consider it best practice when using **R** to annotate heavily!

10.3.1 Reading Data Into R

```
getwd()
```

R is always pointed to a directory on your computer. To find out which directory it is pointed to, run this "get working directory" function. We will assume that we need to change the working directory, and will use the next line of code to set the working directory to the desired path.

```
setwd("E:/FolderName")
```

To set the working directory, change what is in quotation marks to your file location. Also, if you are copying the directory name, it will copy in slashes. You will need to change the slash (i.e., \) to a forward slash (i.e., /). Note that you need your destination name within quotation marks in the parentheses.

```
Ch10_kidspets <- read.csv("Ch10_kidspets.csv")
```

The *read.csv* function reads our data into **R**. What's to the left of the <- will be what the data will be called in **R**. In this example, we're calling the R dataframe "Ch10_kidspets." What's to the right of the <- tells **R** to find this particular .csv file. In this example, our file is called "Ch10_kidspets.csv." Make sure the extension (i.e., .csv) is included in your script. Also note that the name of your file should be in quotation marks within the parentheses.

```
names(Ch10_kidspets)
```

The *names* function will produce a list of variable names for each dataframe as follows. This is a good check to make sure your data have been read in correctly.

```
[1] "Children" "Pets"
View(Ch10_kidspets)
```

The *View* function will let you view the dataset in spreadsheet format in RStudio.

```
summary(Ch10_kidspets)
```

The *summary* function will produce basic descriptive statistics on all the variables in your dataframe. This is a great way to quickly check to see if the data have been read in correctly and to get a feel for your data, if you haven't already. The output from the summary statement for this dataframe looks like this.

```
  Children        Pets
Min.   :1    Min.   : 2
1st Qu.:2    1st Qu.: 4
Median :3    Median : 6
Mean   :3    Mean   : 6
3rd Qu.:4    3rd Qu.: 8
Max.   :5    Max.   :10
```

FIGURE 10.13
Reading data into **R**.

10.3.2 Generating Correlation Coefficients

```
cor.test(Ch10_kidspets$Children, Ch10_kidspets$Pets,
use = "everything",
method = "pearson",
conf.level = 0.95)
```

The *cor.test* function will compute a Pearson (i.e., *method = "pearson"*) correlation coefficient for variables children and pets (i.e., "Ch10_kidspets$Children," "Ch10_kidspets$Pets") and related *p* value using an alpha of .05 (i.e., *conf.level = .95*). The *use = "everything"* command will compute the correlation using all available data ("NA" will be the output if any variables have missing data; we could have used *complete.obs* for listwise deletion or *pairwise.complete.obs* for pairwise deletion, among other options). Because we have no missing data, the method for "use" will not matter; however, if you have missing data, be thoughtful in how you approach this!

Note: To compute Spearman or Kendall's tau, simply change the method (i.e., *method = "pearson"*) to *method = "kendall"* or *method = "spearman,"* using all lowercase letters.

Our output looks like this. We see our observed probability, $p = .037$, which is statistically significant at alpha of .05. We have a 95% confidence interval of the correlation coefficient (.086, .993), and our Pearson correlation coefficient, *r*, is .90.

```
Pearson's product-moment correlation
data: Ch10_kidspets$Children and Ch10_kidspets$Pets
t = 3.5762, df = 3, p-value = 0.03739
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
0.08610194 0.99343752
sample estimates:
cor
0.9
```

FIGURE 10.14
Generating correlation coefficients in **R**.

10.4 Data Screening

As noted previously, the assumptions of the Pearson correlation coefficient are linearity and independence. While the assumption of independence is based on how the data are sampled (with random sampling meeting the assumption of independence), we can use our data to examine the extent to which we meet the assumption of linearity.

10.4.1 Scatterplots to Examine Linearity Using SPSS

Step 1. As alluded to earlier in the chapter, understanding the extent to which linearity is a reasonable assumption is an important first step prior to computing a Pearson correlation coefficient. To generate a scatterplot, go to "Graphs" in the top pulldown menu. From there, select "Legacy Dialogs," then "Scatter/Dot" (see the screenshot for Step 1 in Figure 10.15).

10.4.3.1 Interpreting Linearity Evidence

Scatterplots are also often examined to determine visual evidence of linearity prior to computing Pearson correlations. Scatterplots are graphs that depict coordinate values of *X* and *Y*. *Linearity is suggested by points that fall in a straight line or relatively straight line*. This line may suggest a *positive relation* (as scores on *X* increase, scores on *Y* increase, and vice versa), a *negative relation* (as scores on *X* increase, scores on *Y* decrease, and vice versa), *little or no relation* (relatively random display of points), or a *polynomial relation* (e.g., curvilinear). In this example, our scatterplot generally suggests evidence of linearity and, more specifically, a positive relationship between number of children and number of pets (see Figure 10.21, generated in **R,** and Figure 10.22, generated in SPSS). Thus, proceeding to compute a bivariate Pearson correlation coefficient is reasonable.

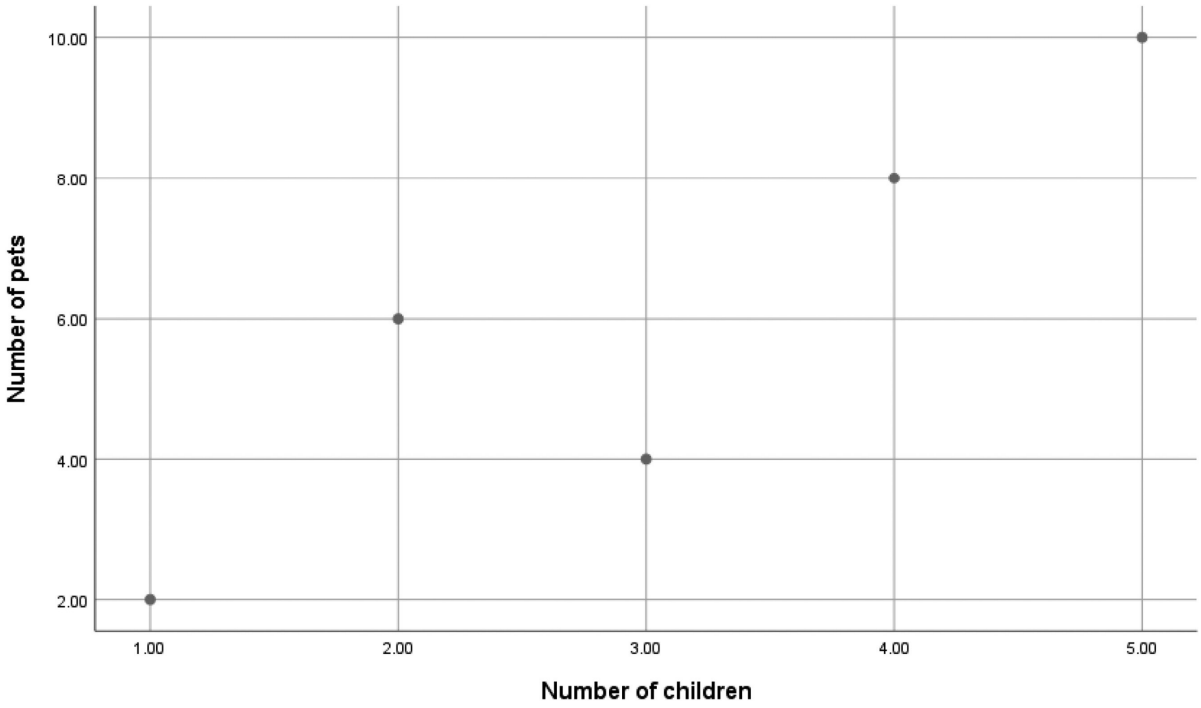


FIGURE 10.22
Scatterplot.

10.5 Power Using G*Power

A priori and post hoc power could again be determined using the specialized software described previously in this text (e.g., G*Power), or you can consult *a priori* power tables (e.g., Cohen, 1988). As an illustration, we use G*Power to compute the post hoc power of our test.

10.5.1.1 Post Hoc Power for the Pearson Bivariate Correlation Using G*Power

The first thing that must be done when using G*Power for computing post hoc power is to select the correct test family. In our case, we conducted a Pearson correlation.

recommended so that you avoid a situation where, post hoc, you find that the sample size was not sufficient to reach the desired level of power (given the observed parameters). In our situation, we don't need to worry that post hoc power was less than desirable as we found a statistically significant correlation.

10.6 Research Question Template and Example Write-Up

Finally, we conclude the chapter with a template and an APA-style paragraph detailing the results from an example dataset. As you may recall, our graduate research assistant, Challie Lenge, was working with the marketing director of the local animal shelter, Dr. Amberly. Challie's task was to assist Dr. Amberly in generating the test of inference to answer her research question, "*Is there a relationship between the number of children in a family and the number of pets?*" A Pearson correlation was the test of inference suggested by Challie. A template for writing a research question for a correlation (regardless of which type of correlation coefficient is computed) follows:

Is there a correlation between [variable 1] and [variable 2]?

It may be helpful to include in the results information on the extent to which the assumptions were met (recall there are two assumptions: independence and linearity). This assists the reader in understanding that you were thorough in data screening prior to conducting the test of inference. Recall that the assumption of independence is met when the cases in our sample have been randomly selected from the population. One or two sentences are usually sufficient to indicate if the assumptions are met. It is also important to address effect size in the write-up. Correlations are unique in that they are already effect size measures, so computing an effect size in addition to the correlation value is not needed. However, it is desirable to interpret the correlation value as an effect size. Effect size is easily interpreted from the correlation coefficient value utilizing Cohen's (1988) subjective standards previously described or in comparison to similar studies that have used like variables. Here is an example paragraph of results for the correlation between number of children and number of pets.

A Pearson correlation coefficient was computed to determine if there is a relationship between the number of children in a family and the number of pets in the family. The test was conducted using an alpha of .05. The null hypothesis was that the relationship would be zero. The assumption of independence was met via random selection. The assumption of linearity was reasonable given a visual review of a scatterplot of the variables along with hypothesis tests to examine quadratic and cubic trends (neither of which were statistically significant).

The Pearson correlation coefficient between children and pets is .90 (*CI* .09, .99), which is positive, is interpreted as a large effect size (Cohen, 1988), and is statistically different from zero ($r = .90$, $n = 5$, $p = .037$). Thus, the null hypothesis that the correlation is zero was rejected at the .05 level of significance. There is a strong, positive correlation between the number of children in a family and the number of pets in the family.

10.7 Additional Resources

This chapter has provided an introduction to conducting correlational analysis. However, there are a number of areas related to correlation, particularly as it relates to correlation as a precursor to regression, that space limitations prevent us from delving into. Those who are interested in general coverage of correlation as a precursor to regression may wish to review Sahay (2016).

Problems

Conceptual Problems

1. The variance of X is 9, the variance of Y is 4, and the covariance between X and Y is 2. What is r_{XY}?

a. .039

b. .056

c. .233

d. .333

2. The standard deviation of X is 20, the standard deviation of Y is 50, and the covariance between X and Y is 30. What is r_{XY}?

a. .030

b. .080

c. .150

d. .200

3. Which of the following correlation coefficients, each obtained from a sample of 1000 children, indicates the *weakest* relationship?

a. −.90

b. −.30

c. +.20

d. +.80

4. Which of the following correlation coefficients, each obtained from a sample of 1000 children, indicates the *strongest* relationship?

a. −.90

b. −.30

c. +.20

d. +.80

5. If the relationship between two variables is linear, which of the following is necessarily true?

a. The relation can be most accurately represented by a straight line.

b. All the points will fall on a curved line.

c. The relationship is best represented by a curved line.

d. All the points must fall exactly on a straight line.